



EEDI

Energy Efficiency Design Index

Engineering the Future – since 1758.

MAN Diesel & Turbo



MAN Diesel & Turbo

Powering the world – responsibly

MAN Diesel & Turbo is the world's leading provider of large-bore diesel engines and turbo-machinery. Our portfolio includes two-stroke and four-stroke engines for marine and stationary applications, turbochargers and propellers, as well as gas and steam turbines, compressors, and chemical reactors.

Our commitment to minimizing fuel consumption while meeting even the most advanced emissions regulations plays a vital role in safeguarding the environment for future generations.

We are seeing unprecedented climate change threaten our vital, yet fragile ecosystems. The knock-on effects of rising temperatures caused by greenhouse gas emissions such as carbon dioxide (CO₂) could have disastrous effects on global agriculture and trade. That's why it has never been more important to limit the cause and effects for future generations.

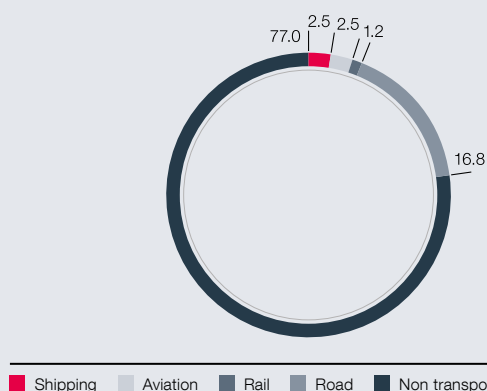
Shipping makes a difference

Maritime transport will continue to expand with increasing globalization, and although shipping already counts as the most efficient form of bulk transportation, the industry has recognized that more can be done. Optimized engines and improved designs lay the foundations for positive change. Working with key stakeholders, the International Maritime Organization (IMO) has outlined new standards for greater efficiency throughout all stages of a ship's lifecycle. One such measure, the Energy Efficiency Design Index (EEDI), is a perfect example of this ambitious goal.

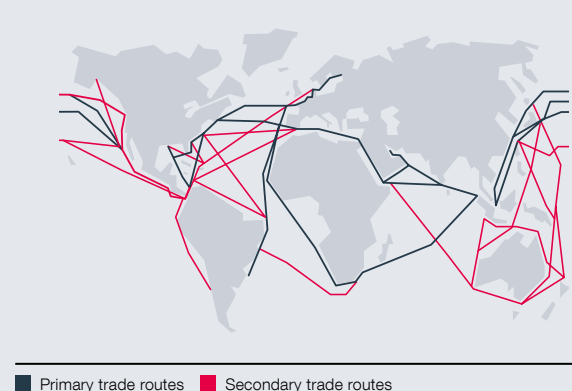
It pays to get on board

With the international shipping industry so committed to ensuring positive change, it will be crucial for individual ship owners and operators to move with the tide. Market-based measures such as levies or emissions trading are foreseeable in the future, and this will only create further incentives to invest in efficient ships. Though final decisions have not yet been made in this respect, the IMO is certainly considering the possibility. So now is the time to act.

Global CO₂ emissions in %



Shipping trade routes



Energy Efficiency Design Index

IMO regulations in a nutshell

What is it?

The EEDI is used to calculate a vessel's energy efficiency. This is based on a complex formula, taking the ship's emissions, capacity, and speed into account. The lower a ship's EEDI, the more energy-efficient it is and the lower its negative impact on the environment. IMO regulations stipulate that ships must meet a minimum energy efficiency requirement, so their EEDI must not exceed a given threshold.

Ships commissioned after January 1, 2013 and weighing 400 GT or more have to meet the requirements. Older vessels are only affected by the EEDI standards if they have undergone a major retrofit in recent years.

That said, ship owners and operators would be wise to consider that EEDI requirements will gradually be tightened: ships built in 2015, 2020 and 2025 will have to meet even higher standards.

the latest updates, RoRo, RoPax, Cruise Ships with diesel-electric propulsion and LNG carriers with diesel-mechanic or diesel-electric propulsion have to meet the limits of the required EEDI. However, based on the results observed in the first phase of the initiative, the IMO intends to expand the EEDI to include additional types of ships in the future. Here too, it will be invaluable for ship operators to keep abreast of the changes.

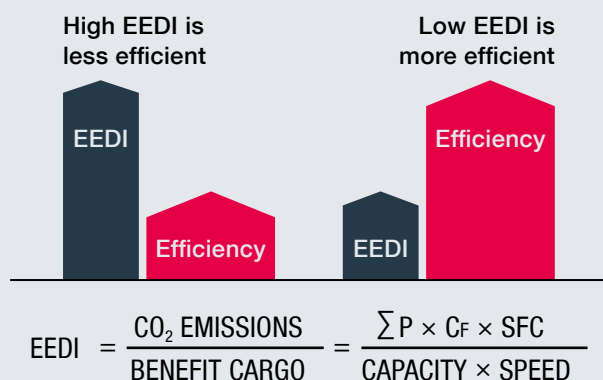
Targeted requirements

At present, the EEDI only applies to the worst offenders when it comes to maritime pollution. In other words: the vessels responsible for the most emissions.

Extension of regulated ship types

Although there were a lot of exemptions in the beginning, the number of ship types to which the EEDI does apply is steadily increasing. According to

High and low EEDI

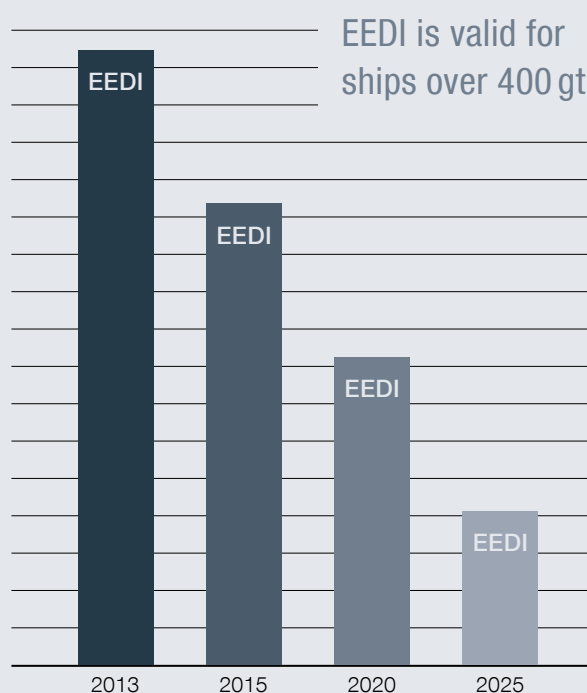


No EEDI for:

- Gas turbine
- Diesel-electric drive*
- Hybrid propulsion
- Offshore

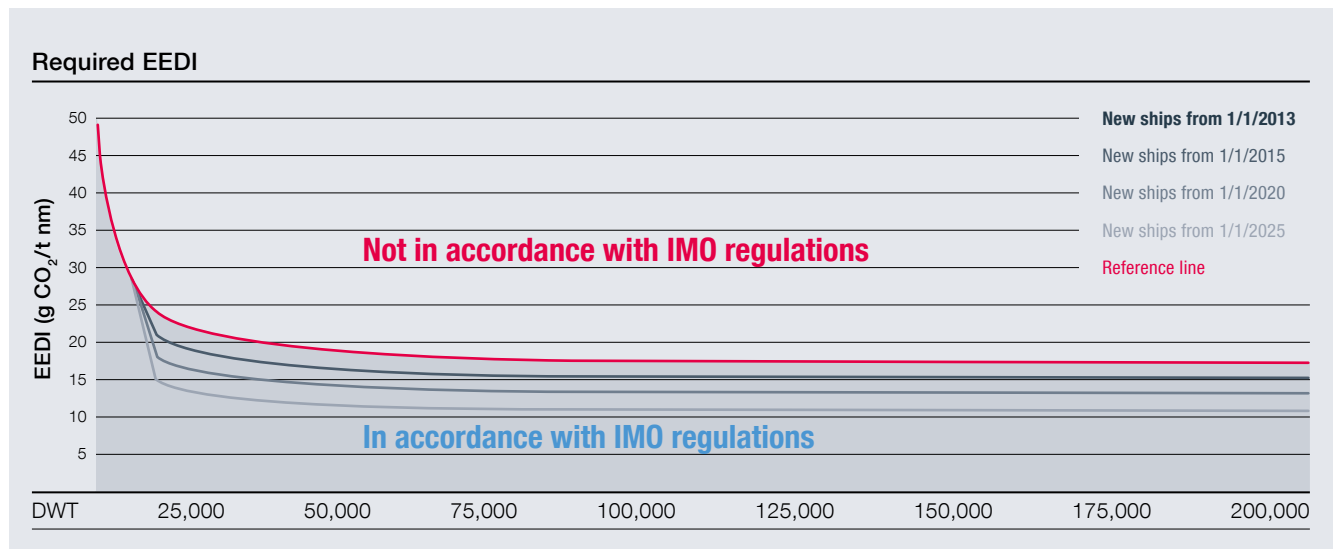
* except for Cruise Passenger Ships and LNG Carriers

New ships over 400 gt (keel-laying after July 2013)



EEDI Spells Efficiency

Definition and application

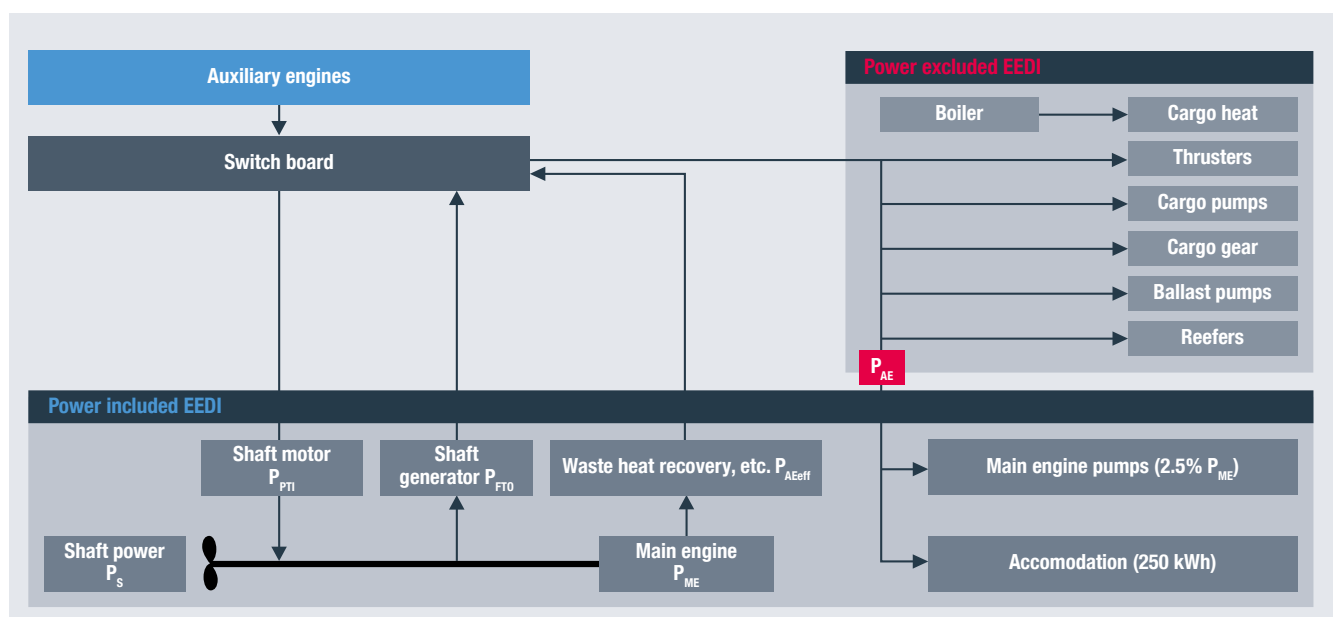


Required EEDI

The required EEDI is the limit for the attained EEDI of a ship and depends on its type and size. Starting with a baseline value in 2013, the limit will be reduced successively in three stages until 2025. The baseline for the required EEDI is a function of the EEDI for vessels built after the year 2000.

Relevant energy consumption

The EEDI assesses the energy consumption of a vessel under normal seafaring conditions, taking into account the energy required for propulsion and the hotel load for the crew. Energy consumed to maintain the cargo and for maneuvering or ballasting is not considered.



EEDI – An Overview

Formula and definitions

MAIN ENGINE EMISSIONS	AUXILIARY ENGINE EMISSIONS	SHAFT GENERATOR/MOTOR EMISSIONS	EFFICIENCY TECHNOLOGIES
$\left(\prod_{j=1}^M f_j \right) \left[\sum_{l=1}^{nME} P_{ME(l)} \cdot SFC_{ME(l)} \cdot C_{FME(l)} \right] + (P_{AE} \cdot C_{FAE} \cdot SFC_{AE}^*) + \left[\left(\prod_{j=1}^M f_j \cdot \sum_{l=1}^{nPTI} P_{PTI(l)} - \sum_{l=1}^{neff} f_{eff(l)} \cdot P_{AEff(l)} \right) C_{FAE} \cdot SFC_{AE} \right] - \left[\sum_{l=1}^{neff} f_{eff(l)} \cdot P_{eff(l)} \cdot C_{FME} \cdot SFC_{ME} \right]$			
$f_l \cdot f_c \cdot f_l \cdot \text{Capacity} \cdot V_{ref} \cdot f_w$ TRANSPORT WORK			

Engine power (P)

Individual engine power depending on application (e.g. PME = 75% Maximum continuous rating for diesel-mechanic propulsion)

$P_{eff(l)}$ Main engine power reduction due to individual technologies for mechanical energy efficiency

$P_{AEff(l)}$ Auxilliary engine power reduction due to individual technologies for electrical energy efficiency

$P_{PTI(l)}$ Power of individual shaft motors divided by the efficiency of shaft generators

P_{AE} Combined installed power of auxilliary engines

$P_{ME(l)}$ Individual power of main engines

CO₂ emissions (C)

CO₂ emission factor based on type of fuel used by given engine

C_{FME} Main engine composite fuel factor

C_{FAE} Auxilliary engine fuel factor

$C_{FME(l)}$ Main engine individual fuel factors

Ship design parameters

V_{ref} Ship speed at reference conditions (see PME definition, etc.)

Capacity Deadweight tonnage (DWT) rating for bulk ships and tankers; a percentage of DWT for container ships; DWT indicates how much can be loaded onto a ship; gross tonnage for passenger ships (cruise)

Specific Fuel Consumption (SFC)

Fuel use per unit of engine power

SFC_{ME} Main engine (composite)

SFC_{AE} Auxilliary engine

SFC_{AE}^* Auxilliary engine (adjusted for shaft generators)

$SFC_{ME(l)}$ Main engine (individual)

Correction and adjustment factors (F)

Non-dimensional factors that were added to the EEDI equation to account for specific existing or anticipated conditions that would otherwise skew the ratings of individual ships

$f_{eff(l)}$ Availability factor of individual energy efficiency technologies (=1.0 if readily available)

f_l Correction factor for ship-specific design elements, e.g. ice-classed ships which require extra weight for thicker hulls

f_w Coefficient indicating the decrease in ship speed due to weather and environmental conditions

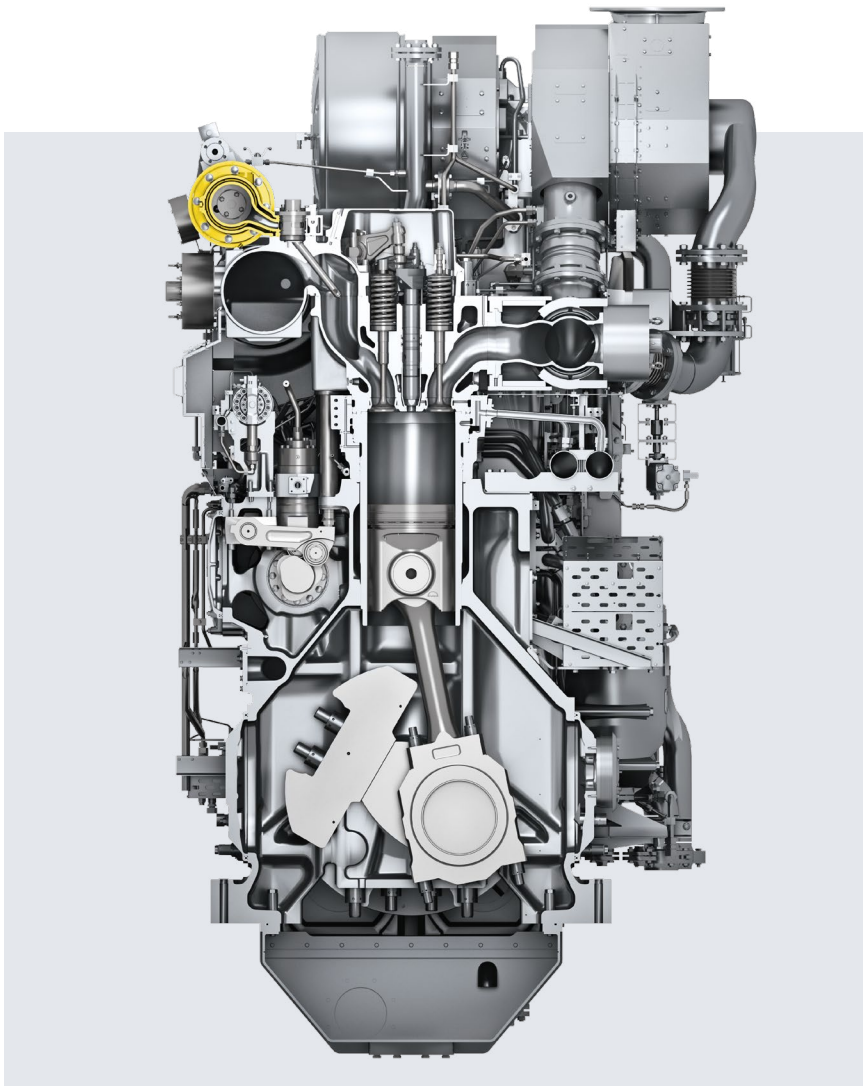
f_l Capacity adjustment factor for any technical/regulatory limitation on capacity (=1.0 if none)

f_c Cubic capacity correction factor (for chemical tankers, LNG carriers and RoPax)

f_l Correction factor to compensate deadweight losses through cargo-related equipment like cranes, RoRo ramps, etc.

Meeting the EEDI Requirements

Optimized engines, components, and engine systems



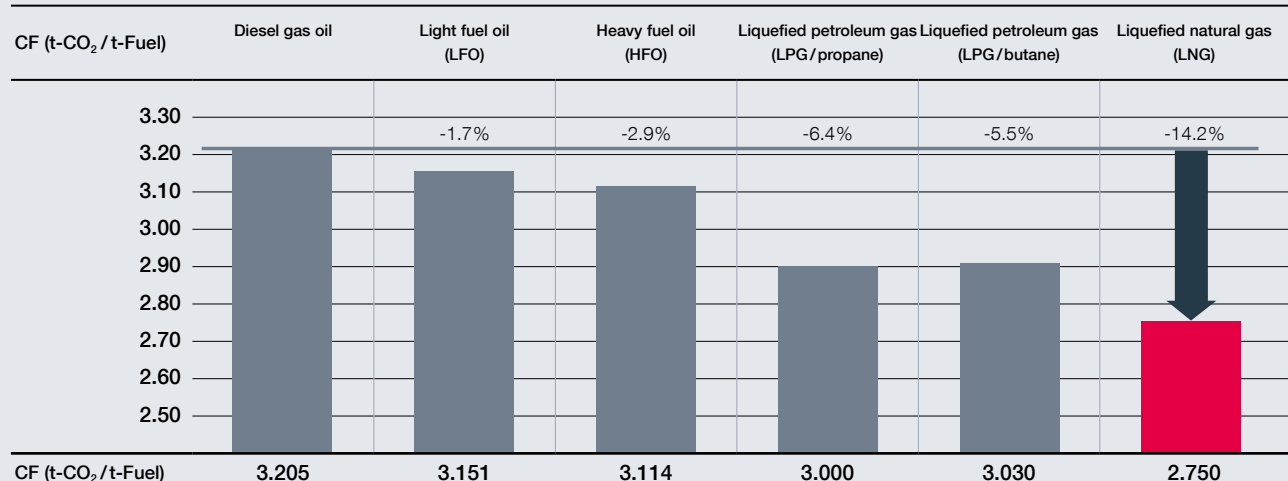
Only ships fitted with state-of-the-art technology will stand a chance of complying with the EEDI. This is where MAN Diesel & Turbo steps in as a competent partner to industry. Our comprehensive range of solutions – including engines, turbochargers and propellers – reflects the high standards that have made us a market leader across the seven seas.

Banking on efficiency

Burning liquefied natural gas (LNG) produces less CO₂ than other conventional sources, making it a powerful alternative in achieving a significantly reduced EEDI. MAN Diesel & Turbo has recently introduced a range of extremely efficient and versatile dual-fuel engines, suitable for almost any type of shipping. With these engine models, ship owners benefit from attractive gas prices and full fuel flexibility.

Due to reduced carbon factors, the use of LNG by the new MAN 35/44DF results in approx. 14% lower EEDI

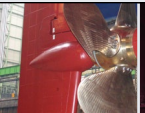
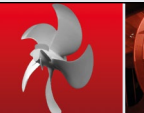
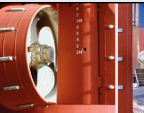
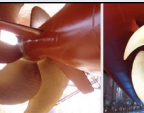
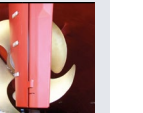
Carbon factors (CF)



Meeting the EEDI Requirements

Optimized engines, components, and engine systems

Various solutions on the market today – and how they can be combined

	Post swirl fins	Rudder bulb	Kappel	PBCF	AHT nozzle	Mewis duct	Pre swirl fins	Efficiency rudders
								
Post swirl fins	2–3%							
Rudder bulb		2–5%						
Kappel			3–5%					
PBCF				2–5%				
AHT nozzle					5–8%			
Mewis duct						3–8%		
Pre swirl fins							3–5%	
Efficiency rudders								2–4%

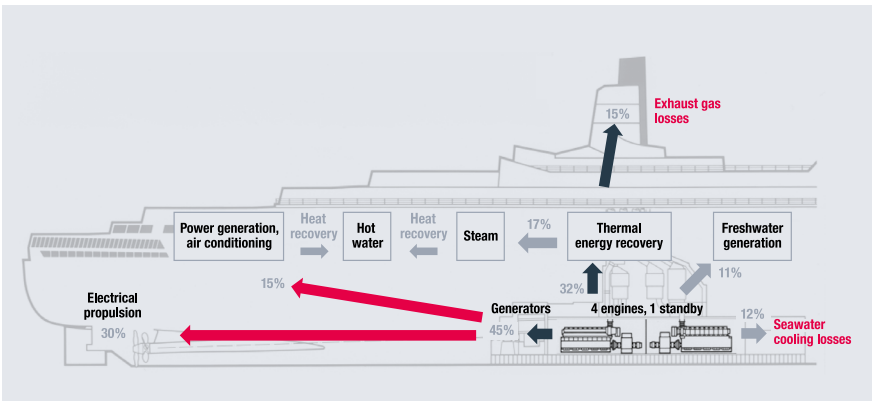
Can be combined

Can sometimes be partially combined

Should not be combined

Clever in combination

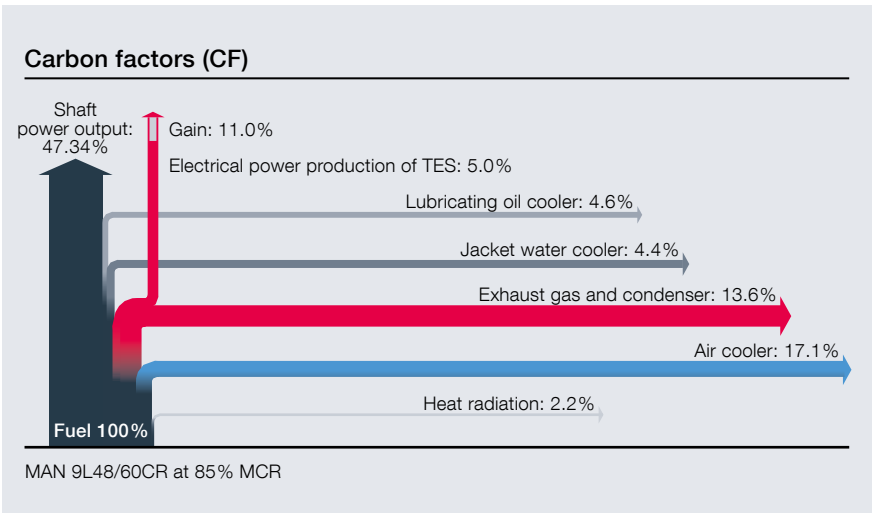
Enhance your vessel with one or more of the many MAN Diesel & Turbo products designed to boost energy efficiency. Take, for example, our robust Kappel propellers: used in combination with a rudder bulb, they contribute to ultimate ship performance. Whether your engine is driven by liquid or gaseous fuels, you can increase engine efficiency by up to 10%.



Typical energy/heat balance of a cruise ship

Less input for greater output

Truly efficient engines have the capacity to derive more power from less fuel. At MAN Diesel & Turbo, we have channeled our expertise into adhering to a simple maxim: less is more. That's why we let nothing go to waste, not even the excess heat produced by the combustion process. With our engine systems, this heat is recovered, providing up to ten percent more power. It can easily be used to run a steam turbine or generator, or can flow into heating for accommodation and cargo.



Efficiency in Action

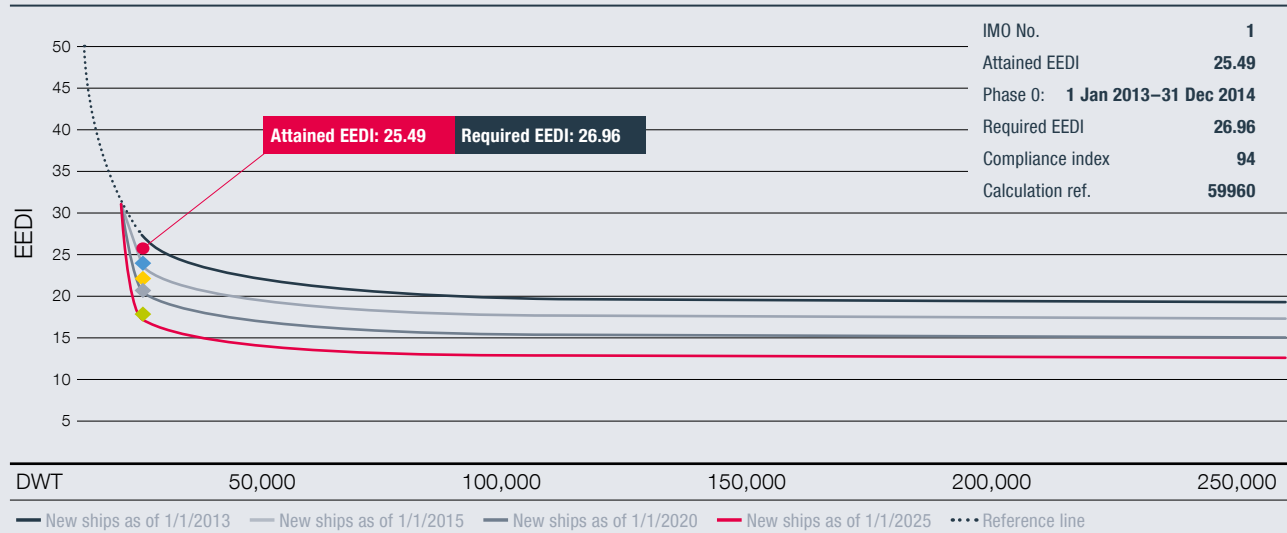
Attained versus required EEDI

Container ship



DWT design draft 15,375 t
 ME 11,200 kW (MAN B&W 8L58/64)
 AE 4 x 1,720 kW (MAN 8L21/31)
 Generator efficiency 93%
 Speed 19 knots
 Diesel/Gas Oil, ISO 8217, DMC – DMX

Requirement container ship



Sensitivity

- **Attained EEDI: 25.49**
- ◆ **Attained EEDI: 22.29**
(with MAN 12V51/60DF)
- ◆ **Attained EEDI: 18.06**
(with engine MAN 9L51/60DF; 9,000 kW and 18 knots)
- ◆ **Attained EEDI: 24.28**
(with engine MAN 7S50ME-B8, two-stroke)
- ◆ **Attained EEDI: 21.06**
(with engine MAN 7S50ME-B8, two-stroke and LNG)

Assumptions and considerations:

All variations are only achieved by changing the main engine characteristics.

Efficiency in Action

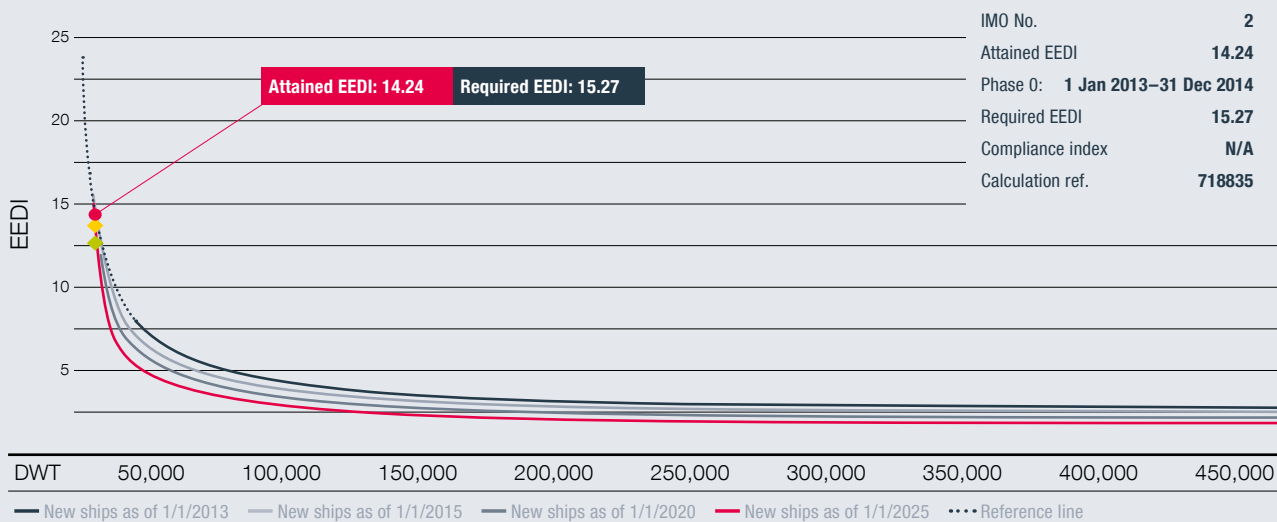
Attained versus required EEDI

Requirement tanker (2008)



DWT design draft 7,900 t
 ME 3,360 kW (MAN 6L32/44CR)
 AE 1 x 1,290 kW (MAN 6L21/31)
 Generator efficiency 93%
 Speed 13.3 knots
 Diesel/Gas Oil, ISO 8217, DMC – DMX

Requirement tanker



Sensitivity

- **Attained EEDI: 14.24**
- ◆ **Attained EEDI: 13.84**
 (with engine MAN 9L27/38; 3,060 kW and 185 g/kWh SFC*)
- ◆ **Attained EEDI: 12.64**
 (using MAN 6L35/44DF)

* 85% MCR

Assumptions and considerations:

All variations are only achieved by changing the main engine characteristics.

All data provided in this document is non-binding. This data serves informational purposes only and is not guaranteed in any way. Depending on the subsequent specific individual projects, the relevant data may be subject to changes and will be assessed and determined individually for each project. This will depend on the particular characteristics of each individual project, especially specific site and operational conditions. Copyright © MAN Diesel & Turbo. D2366498EN-N3 Printed in Germany GGKM-AUG-07162

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